

Taylor Pentz

Senior NLP / LLM Application Engineering

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Professional Summary

Senior NLP / LLM Application Engineer with 11+ years in software engineering across SaaS, fintech, healthcare, analytics, and enterprise workflow products and recent delivery using Python, NLP, transformers, RAG, cloud services, databases, automated testing, and CI/CD; modernizes legacy systems, fixes production reliability issues, improves performance, mentors engineers, and turns complex requirements into secure, scalable products with measurable business impact.

Skills

- **Primary Stack:** Python, NLP, transformers, RAG, prompt evaluation, FastAPI; practical delivery across SaaS, fintech, healthcare, analytics, and enterprise workflow products.
- **AI Depth:** RAG design, embeddings, prompt workflows, model evaluation, latency/cost optimization, retrieval quality, guardrails, feedback loops, and AI observability.
- **Architecture:** RAG architecture, AI agents, model serving, feature pipelines, human-in-the-loop review; API boundaries, domain modeling, secure design, scalability reviews, and production-readiness planning.
- **Cloud and DevOps:** AWS, Azure OpenAI, GCP, Docker, Kubernetes, GitHub Actions; Docker, CI/CD pipelines, release automation, feature flags, rollback plans, and environment governance.
- **Databases and Messaging:** PostgreSQL, pgvector, Pinecone, Weaviate, Redis, Snowflake; schema design, indexing, query tuning, migrations, caching, reconciliation, and reliable async processing.
- **Testing and Quality:** Pytest, evaluation datasets, regression prompts, model drift checks, API tests; unit, integration, contract, end-to-end, regression, load, and smoke testing aligned with release risk.
- **Monitoring and Security:** OpenTelemetry, Datadog, CloudWatch, ELK, Sentry, structured logs, alerts, OWASP, OAuth, RBAC, incident response, and RCA.
- **Senior Practices:** technical leadership, code reviews, architecture reviews, mentoring, Agile planning, stakeholder communication, legacy modernization, and production support.

Work Experience

SumatoSoft

Senior Software Engineer

Boston, MA Remote

Apr 2023 - Dec 2025

- Led AI-enabled product delivery for **internal knowledge and automation workflows** using **Python, NLP, transformers, RAG, prompt evaluation, FastAPI**, centered on **Python 3.10+**, **FastAPI**, **vector search**, and **current LLM APIs**, connecting retrieval, model APIs, evaluation data, user feedback, and secure application services.
- Modernized legacy modules with duplicated validation, inconsistent state handling, and fragile release paths into **typed, testable, reusable components and services**, reducing production defects by **32%** across high-change customer workflows.
- Resolved hallucination, stale context, high latency, and inconsistent answers by improving **chunking, metadata filters, prompt templates, vector search, caching, model routing, and evaluation datasets**.
- Designed secure integration boundaries with **OAuth, RBAC, tenant-aware filters, OpenAPI contracts, audit logging, and defensive validation**, preventing cross-account data exposure during partner and administrator workflows.
- Implemented reliable asynchronous processing with **idempotency keys, retry policies, dead-letter handling, reconciliation jobs, and transactional persistence**, eliminating duplicate business actions after provider timeouts.
- Moved fragile deployments toward **Docker-based CI/CD, infrastructure checks, feature flags, environment promotion controls, and repeatable rollback procedures**, improving release confidence for distributed teams.
- Added automated AI quality gates with **golden datasets, regression prompts, retrieval checks, toxicity and privacy reviews, API tests, and output scoring**, preventing prompt changes from degrading customer workflows.
- Built observability with **OpenTelemetry, structured logs, Datadog and CloudWatch alerts, traces, dashboards, and incident runbooks**, accelerating diagnosis of latency spikes, memory growth, failed jobs, and webhook failures.
- Mentored engineers through **architecture reviews, pair debugging, pull-request standards, incident retrospectives, and performance-focused refactoring**, raising technical consistency across frontend, backend, QA, DevOps, and product groups.

Hexaview Technologies Inc.

Software Engineer

New York, NY

Jul 2018 - Mar 2023

- Delivered client-facing fintech and analytics features using **Python, scikit-learn, PyTorch, Pandas, NLP pipelines, feature services, model APIs, MLflow, and monitoring dashboards**, centered on **Python 3.7-3.9, scikit-learn, PyTorch, Pandas, and MLflow**, with **prompt engineering, embeddings, model evaluation, data pipelines, observability, secure APIs**, supporting customer dashboards, reporting workflows, account servicing, and external data integrations.
- Refactored older screens, services, and shared libraries that mixed business rules with presentation logic, improving maintainability and reducing regression bug leakage by **29%** across three major release cycles.
- Built service and API integrations for payments, CRM, identity, notifications, and reporting providers, handling **duplicate callbacks, schema drift, partial outages, retries, and reconciliation** without losing customer activity.
- Fixed production memory and worker stability issues by profiling retained objects, closing database cursors, limiting batch size, and introducing safer queue-processing controls for long-running jobs.
- Optimized reporting and search workloads through **composite indexes, query rewrites, cached summaries, asynchronous generation, and defensive timeouts**, stabilizing response times as customer records grew.
- Migrated older modules toward modern runtime, dependency, and build standards while preserving backward-compatible API contracts for dependent clients and scheduled enterprise releases.
- Expanded automated validation with **component tests, API tests, contract fixtures, browser tests, and release smoke suites**, reducing manual QA bottlenecks and catching provider response changes earlier.
- Partnered with architects, product managers, QA, and client stakeholders to split large modernization initiatives into smaller releases with measurable acceptance criteria, release notes, and support documentation.

SimbirSoft Company

Software Developer

Boston, MA

Mar 2015 - Jun 2018

- Built enterprise workflow, e-commerce, and internal operations modules using **Python, scikit-learn, Pandas, NumPy, basic NLP, search relevance, classification scripts, SQL datasets, and REST APIs** with **SQL databases, REST APIs, queues, and CI builds**, supporting approvals, orders, notifications, imports, and reporting.
- Replaced brittle server-rendered and script-heavy flows with cleaner application modules, improving task completion speed by **24%** for operations users working through repetitive approval and data-entry processes.
- Implemented service-layer validation, transaction boundaries, and database constraints that corrected duplicate submissions, inconsistent statuses, and race conditions caused by browser retries or concurrent updates.
- Assisted modernization from manual deployments and tightly coupled code into **repeatable builds, environment profiles, automated tests, structured logging, and safer release packages** across staging and production.
- Improved slow database-backed screens through **index reviews, query tuning, pagination, selective caching, and explain-plan analysis**, reducing support complaints tied to large customer and catalog tables.
- Created REST endpoints and background jobs for notifications, file imports, invoice synchronization, and status tracking while adding error queues and replay scripts for failed integrations.
- Worked with QA, analysts, and senior engineers to clarify requirements, investigate production defects, write integration tests, and document root causes for recurring customer-impacting issues.

Microsoft

Software Developer Intern

Cambridge, MA

Aug 2014 - Feb 2015

- Contributed to internal developer productivity and reporting tools using **Python scripts, C#, ASP.NET MVC/Web API, JavaScript, TypeScript, SQL Server, and internal Azure-hosted tools**, supporting SQL-backed services, administrative dashboards, validation rules, data exports, and workflow automation for engineering teams.
- Implemented searchable grids, export utilities, form validation, and role-aware visibility rules that helped internal users review operational data and complete routine support tasks more efficiently.
- Investigated defects in data access, validation, and workflow-state handling by reproducing failures, comparing expected outcomes, and preparing fixes reviewed by senior engineers before merge.
- Added unit and integration tests around controller logic, utility classes, SQL queries, and edge-case validation, strengthening regression coverage for tools that had relied heavily on manual checking.
- Participated in code reviews, daily standups, release readiness checks, and design discussions, gaining production-minded engineering habits around maintainability, reliability, and secure implementation.
- Documented setup steps, API assumptions, troubleshooting notes, and known limitations so future interns and full-time engineers could onboard faster and maintain the internal tools confidently.

Selected Projects

Enterprise AI Knowledge Assistant

- Built an AI knowledge workflow using **Python, NLP, transformers, RAG, prompt evaluation, FastAPI**, retrieval pipelines, embeddings, prompt templates, secure APIs, feedback capture, and evaluation datasets for internal support and customer operations.

RAG Evaluation and Model Quality Platform

- Implemented AI quality controls with golden-answer tests, retrieval scoring, prompt regression checks, latency tracking, cost monitoring, and human review loops for safer production rollout.

Certifications

- **AWS Certified Machine Learning - Specialty**
- **Microsoft Certified: Azure AI Engineer Associate**
- **Databricks Certified Machine Learning Associate**
- **AWS Certified Developer - Associate**
- **Professional Scrum Developer I (PSD I)**

Education

University of North Carolina at Chapel Hill

Bachelor's Degree in Computer Science, 2010 - 2014